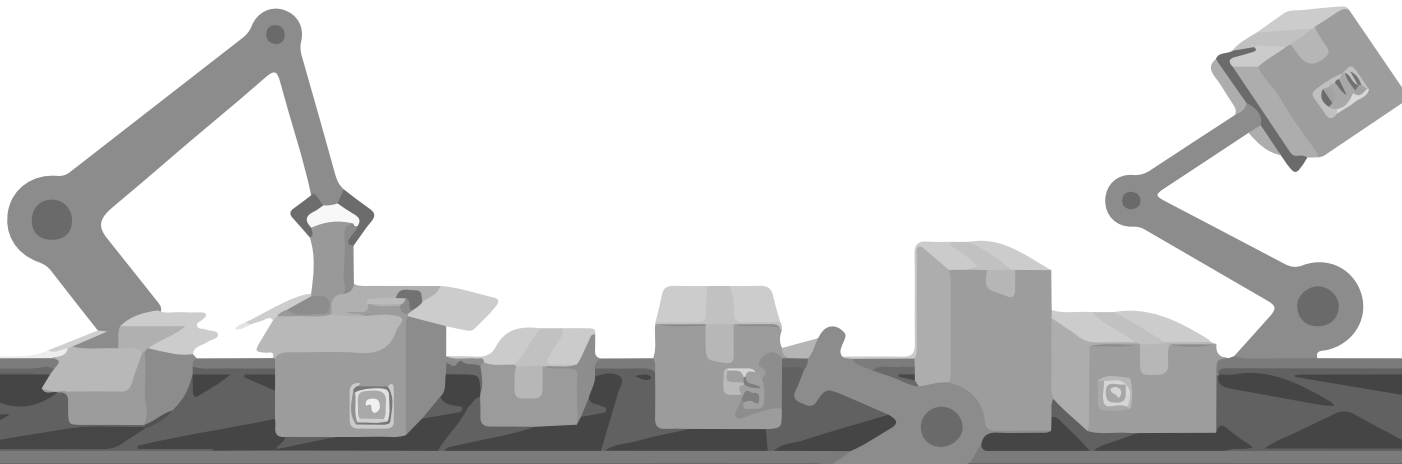
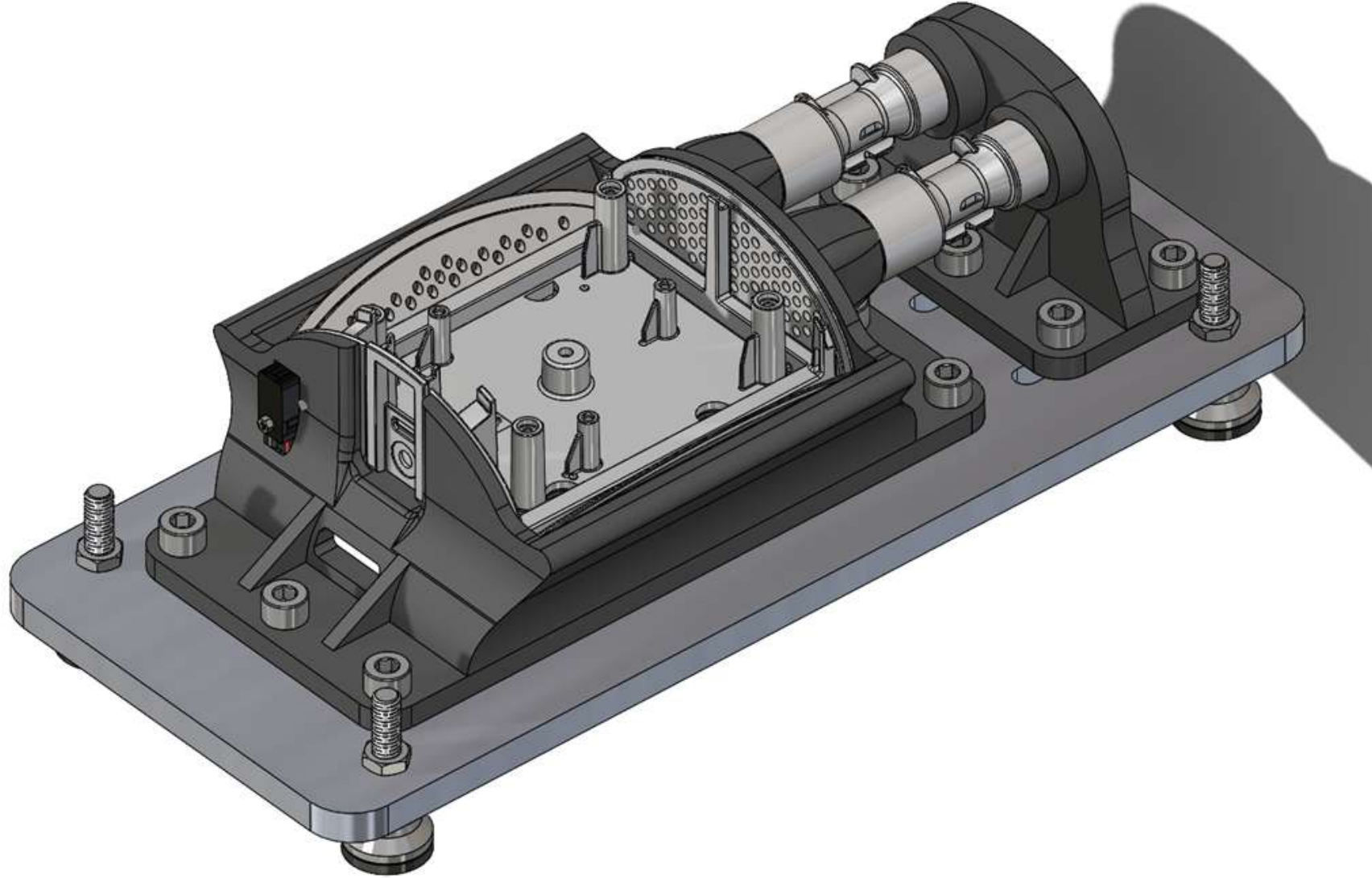


Process/Industrial Engineering

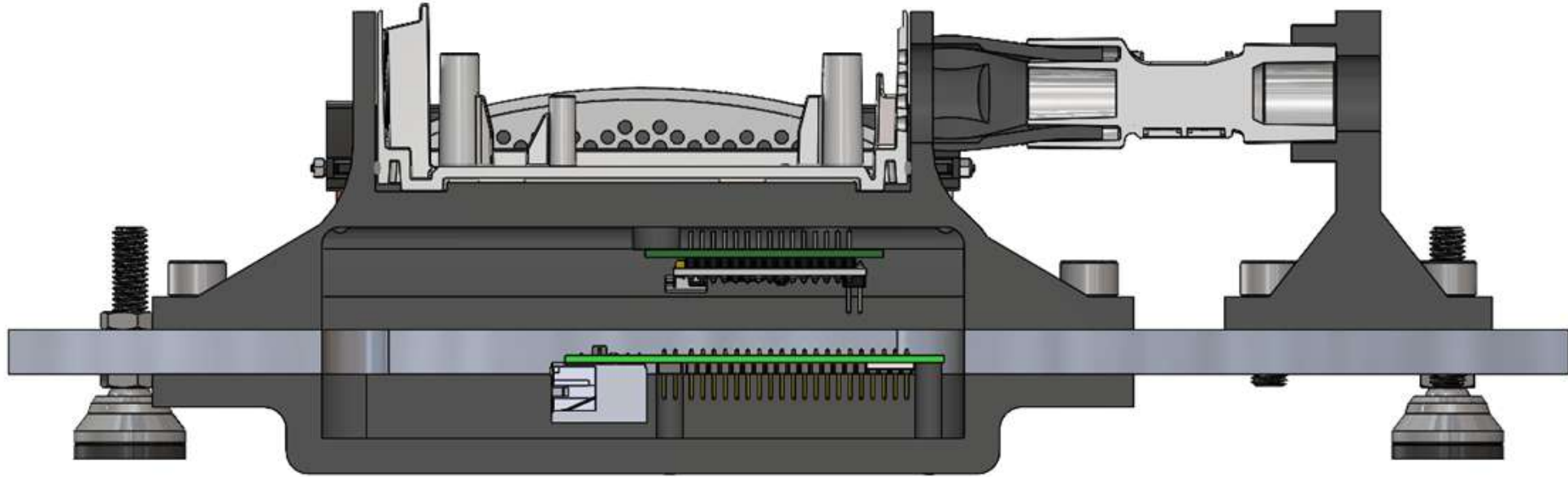
Spenser Millburn – Portfolio 2019-2021



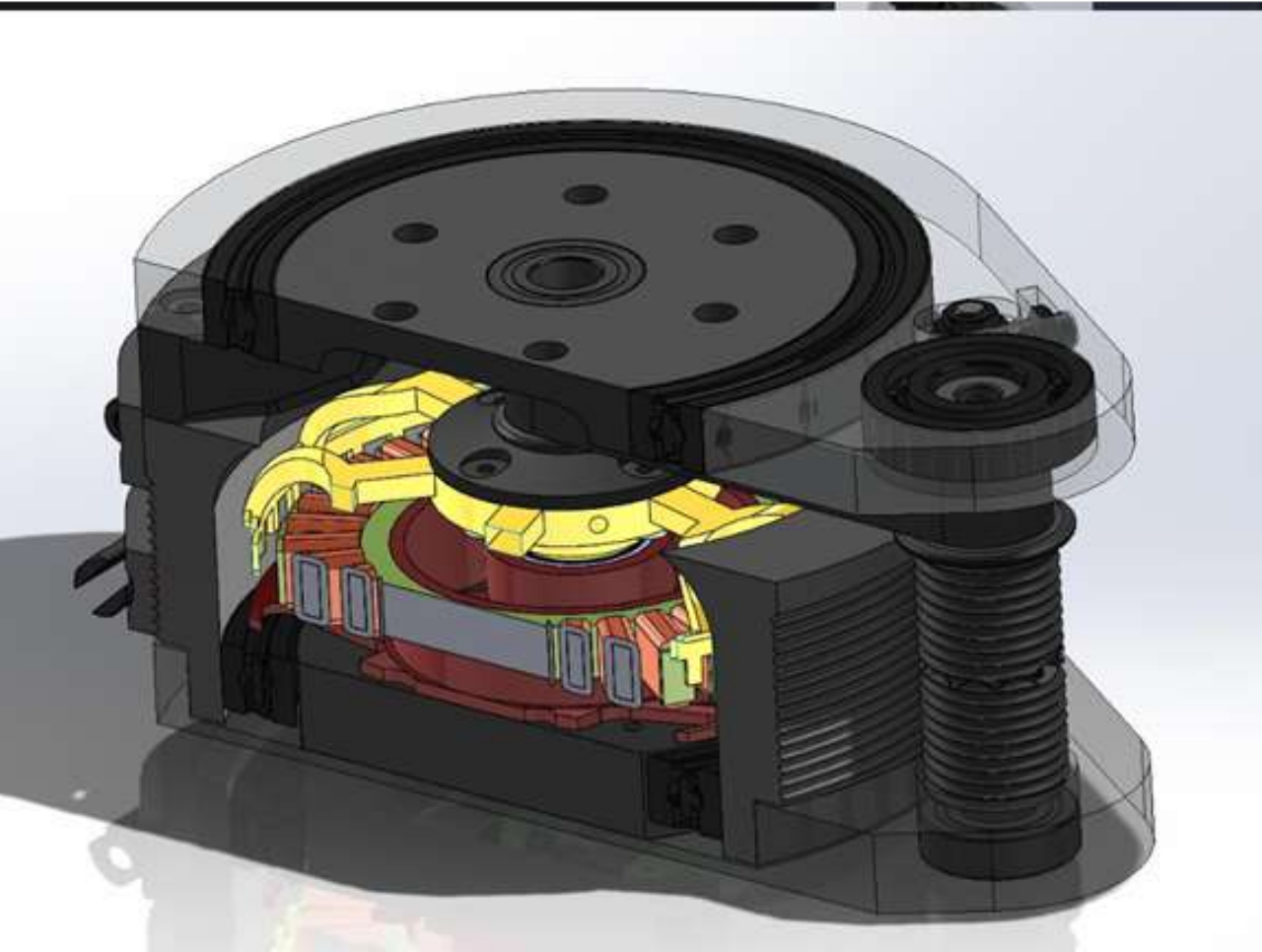
Production Fixture: High Throughput Airflow Validation



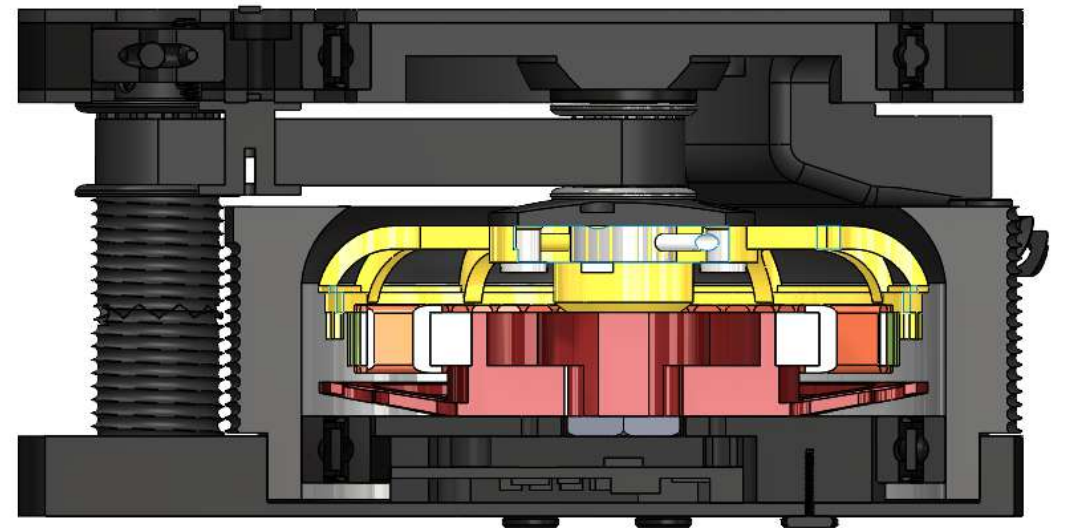
Production Fixture: High Throughput Airflow Validation



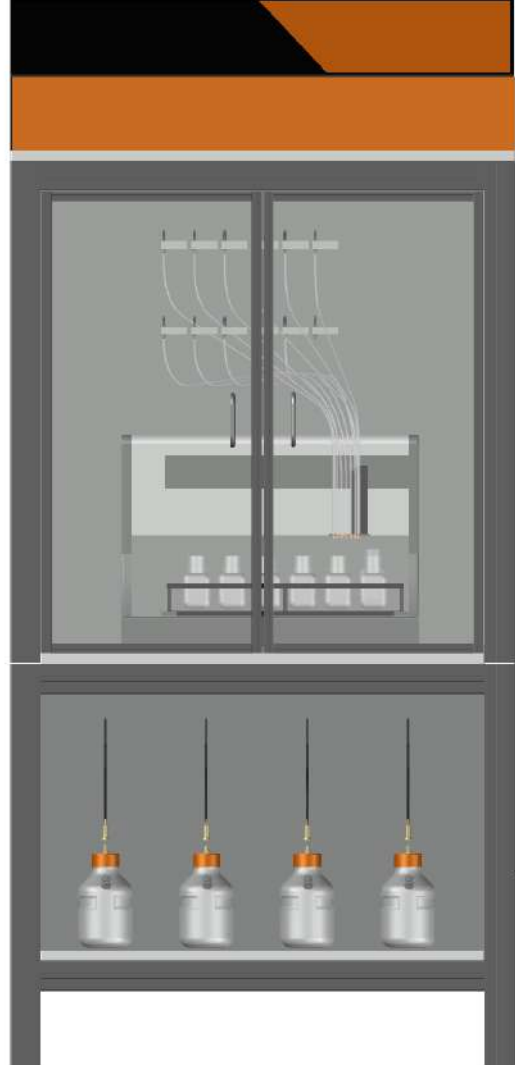
Quadruped Robot: Capstan Drive



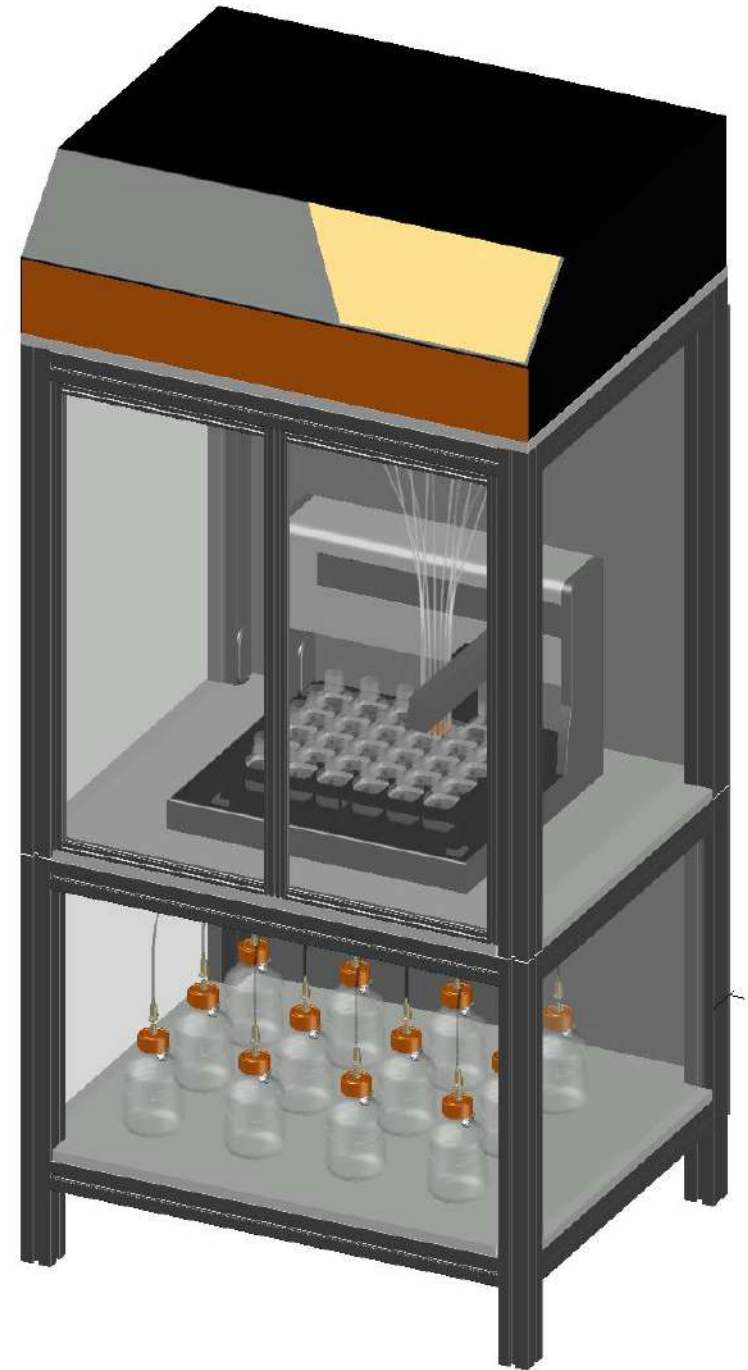
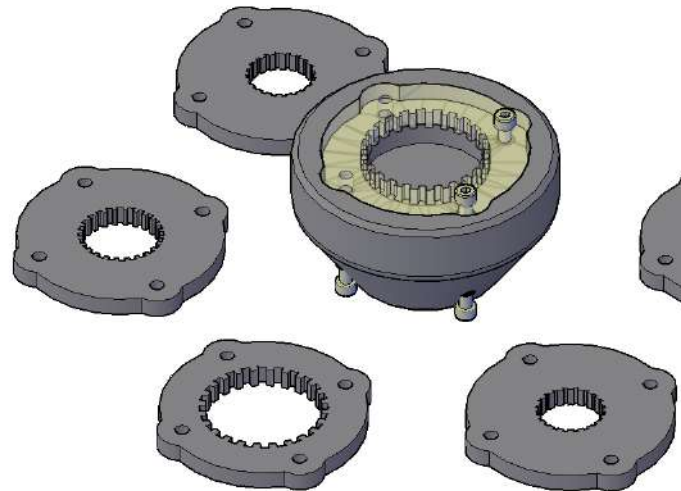
- Cable driven gear reduction
- Brushless motor successfully mounted internally in the large drum.
- Less than \$200 in parts
- Parametric model in process for easy reconfiguration of gear ratios.



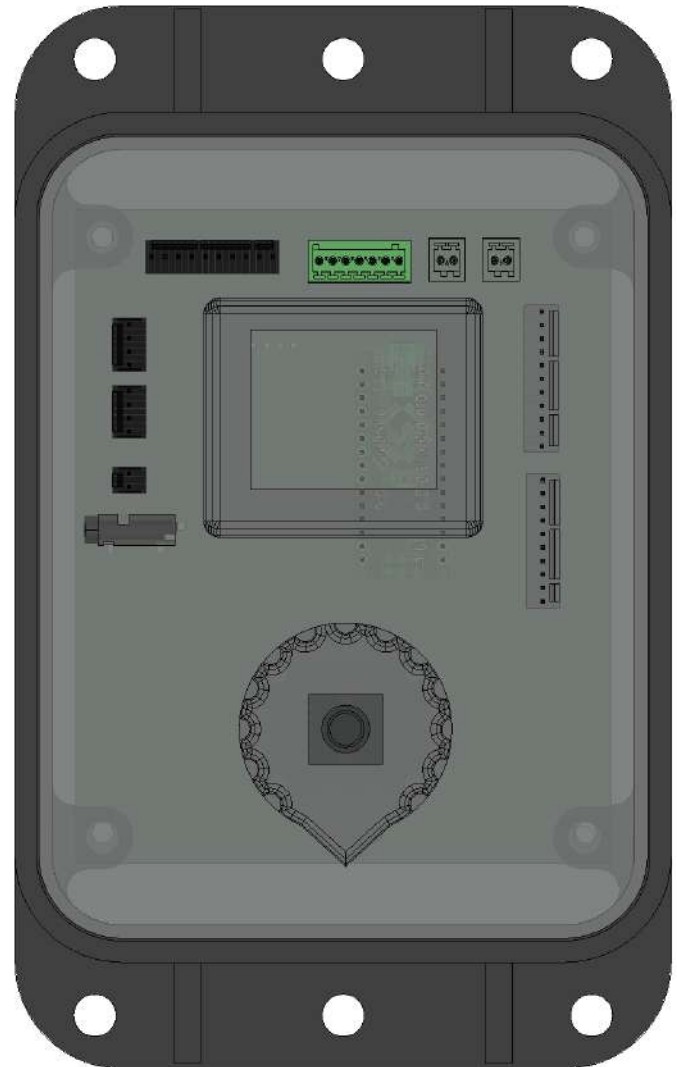
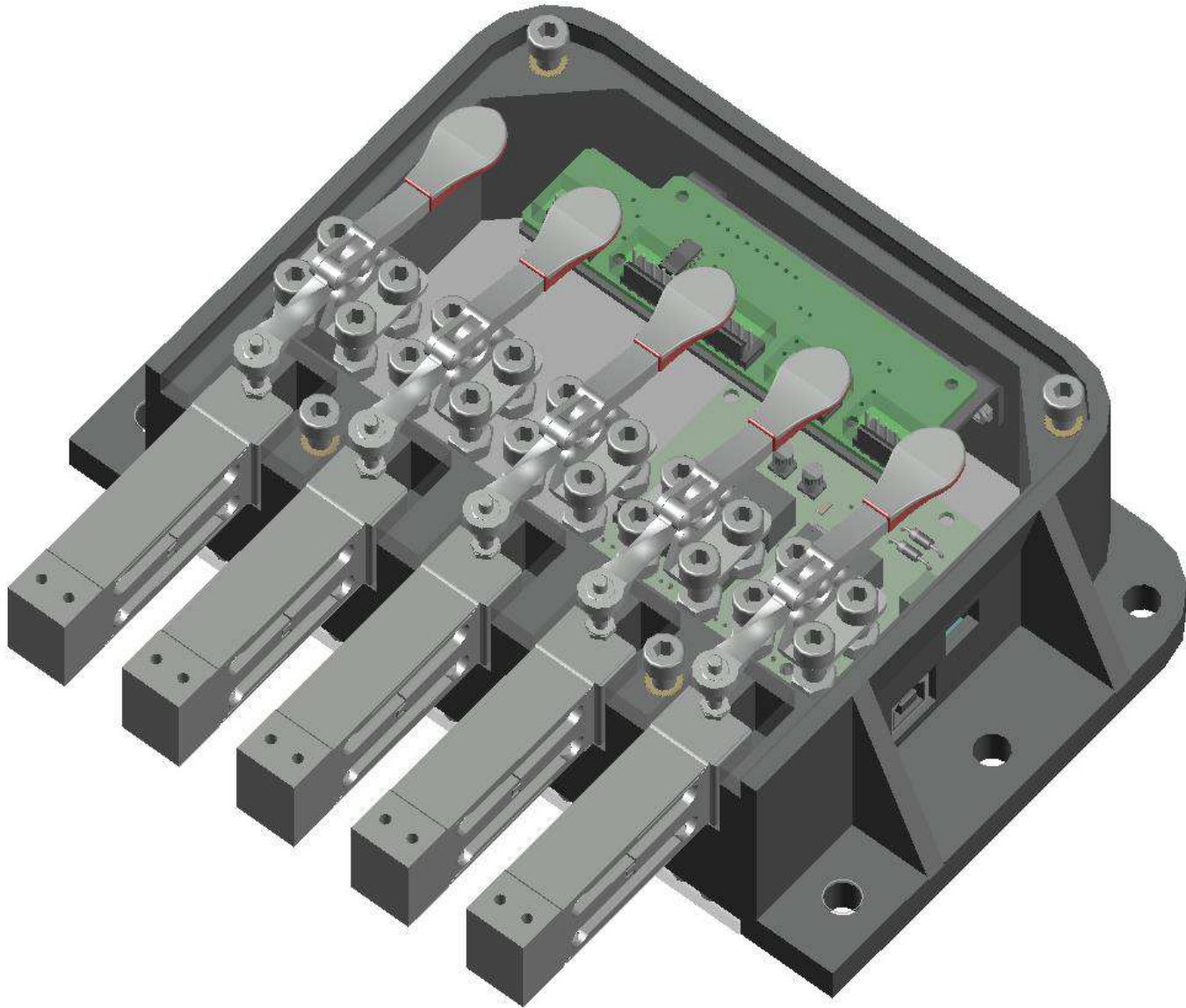
Reagent Lab Automation: Liquid Handler/Capper



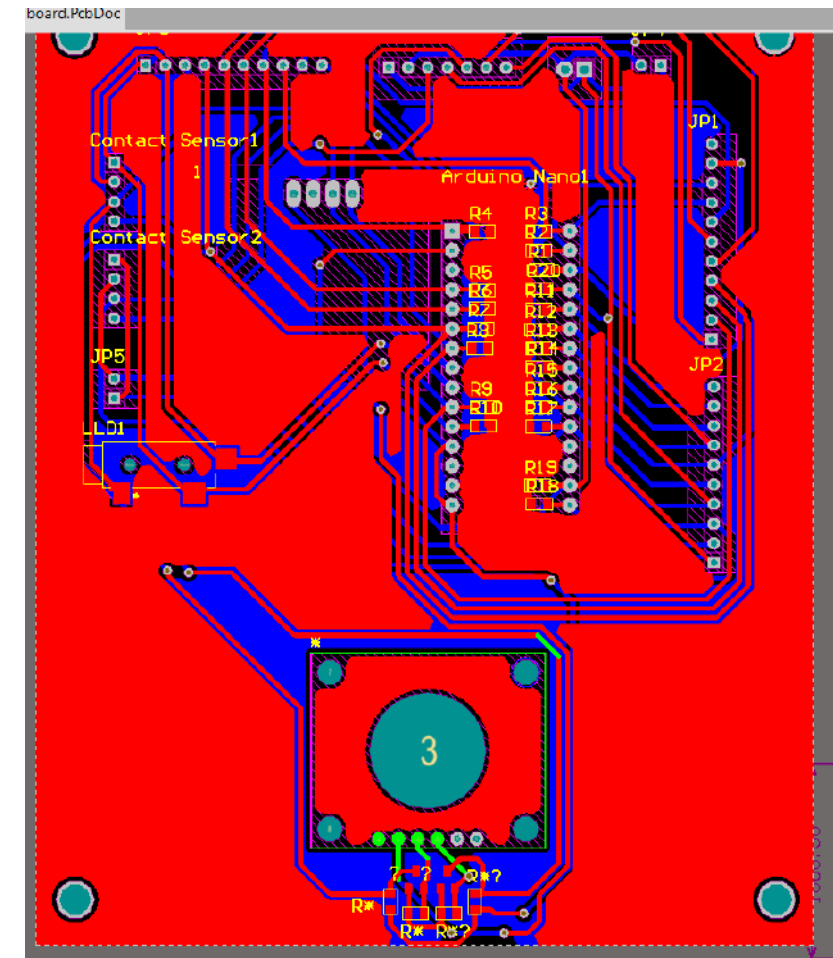
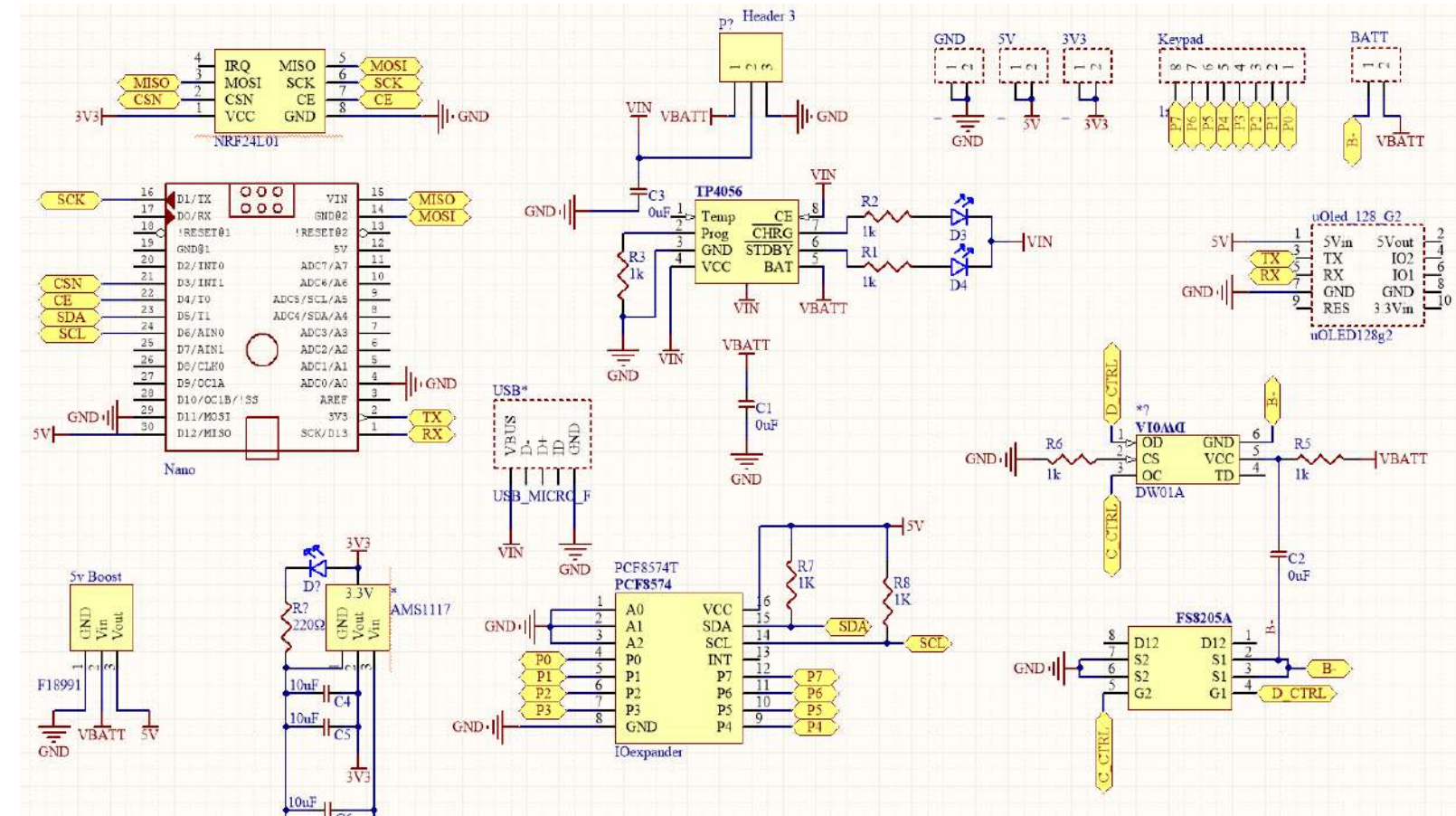
- Based on GX 271 liquid handler
- CO2 dispensing system
 - **Solenoid Valves** controlled by microcontroller/microprocessor
- Custom capping tool
- Laser cut trays to secure bottles



Hardware: Test Stations



Cable Test Station : PCB Design

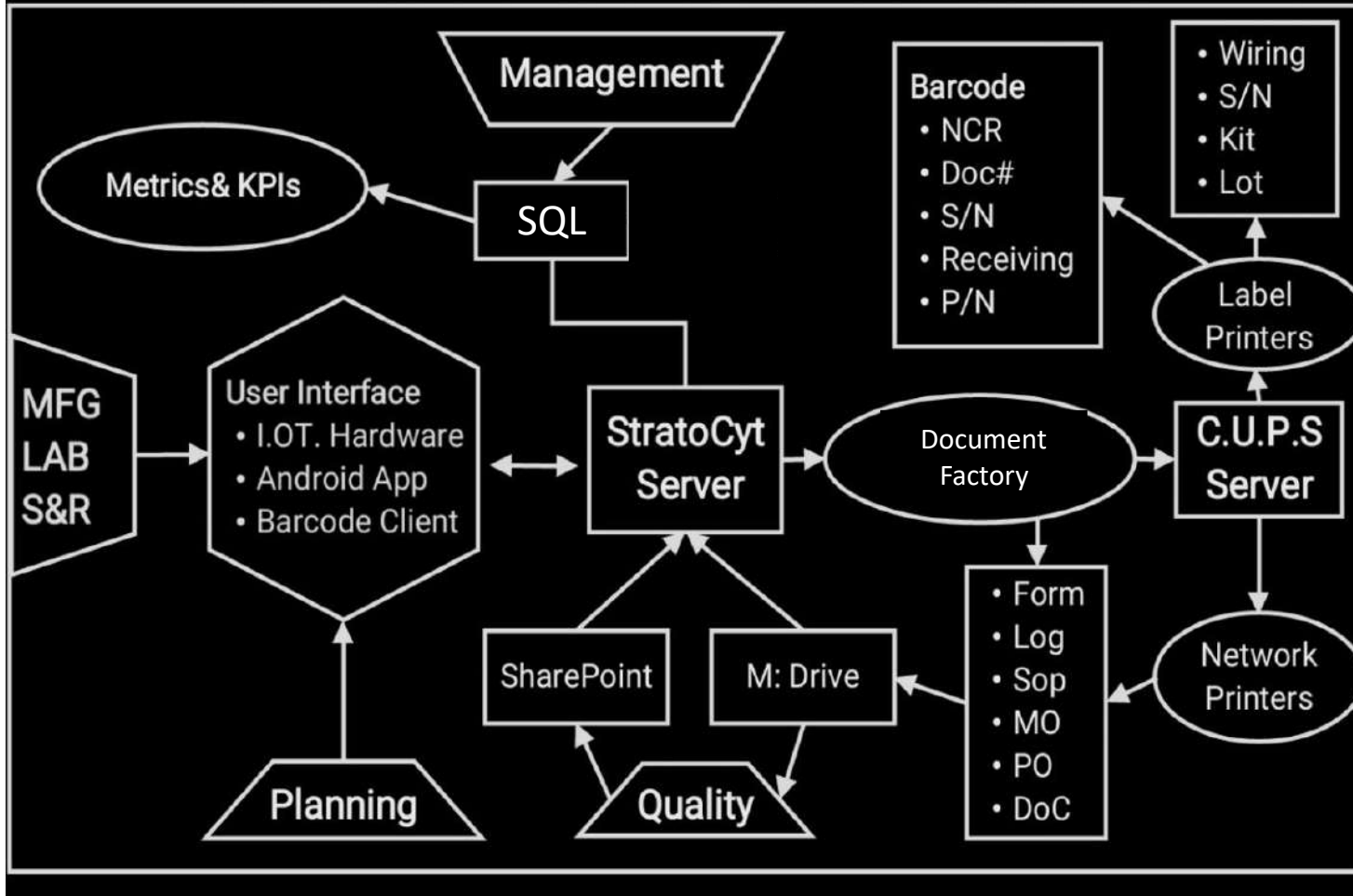


PCB : Rapid Prototyping, Hand Etching



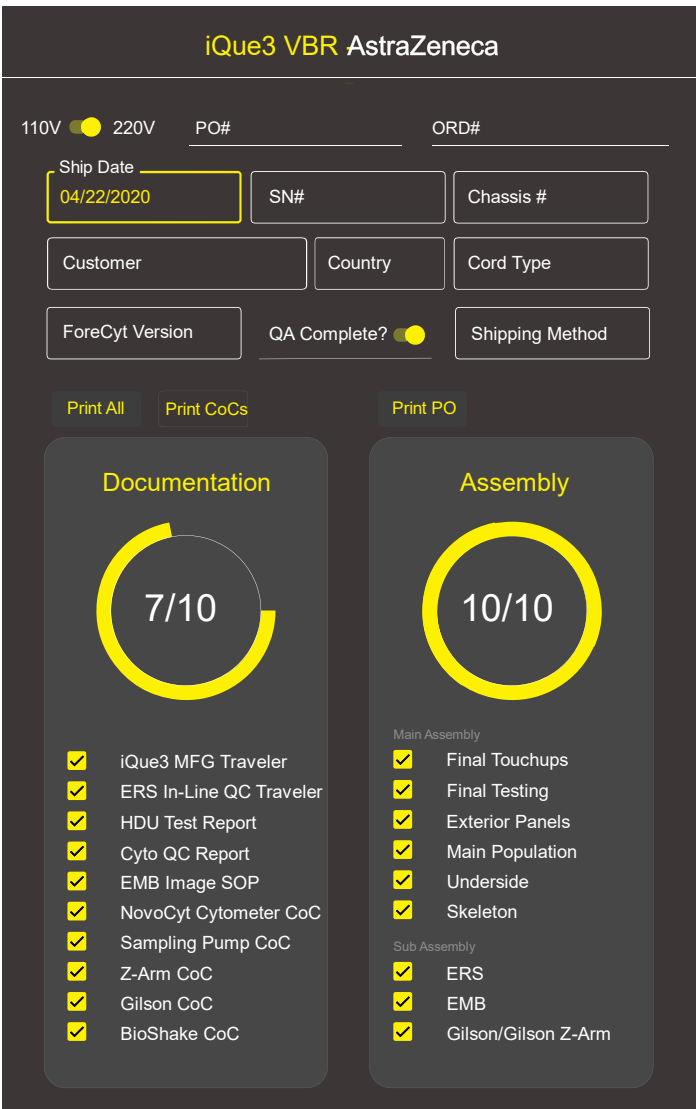
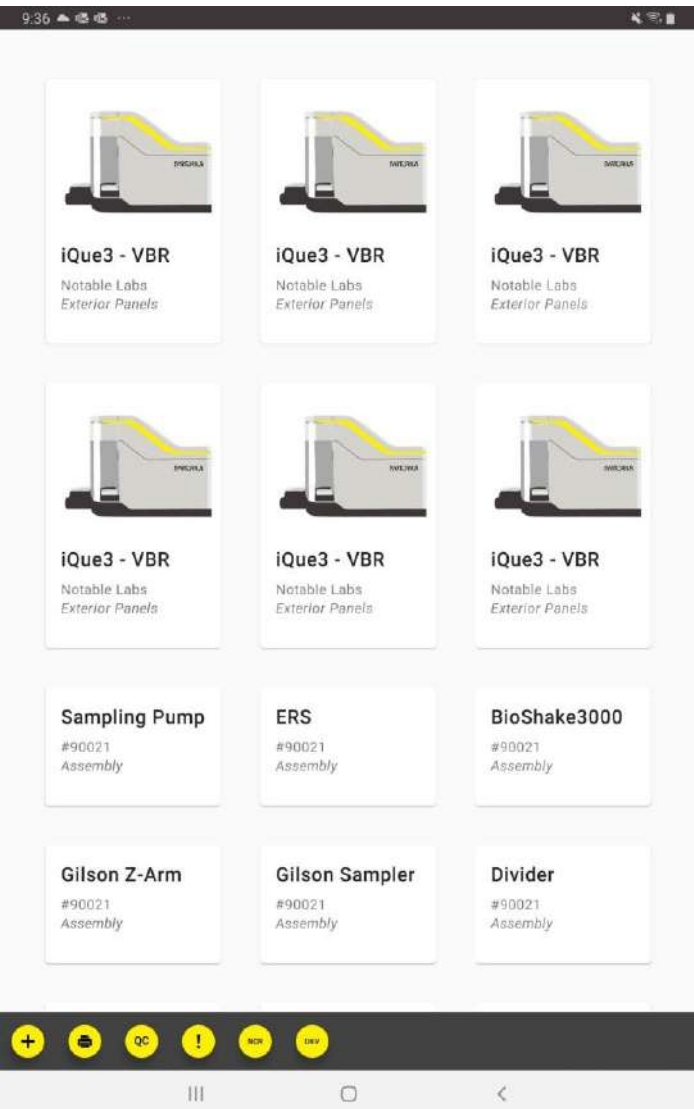
StratoCyt – Custom Automation Server

Reference Diagram



- Microsoft SQL Server Database
 - Stores essential production data for real time metrics
- Python Socket Server
- Automatic document and label generation
 - Automates paperwork system
- Mobile front-end for efficient in process user interface

StratoCyt: Front end: Android app

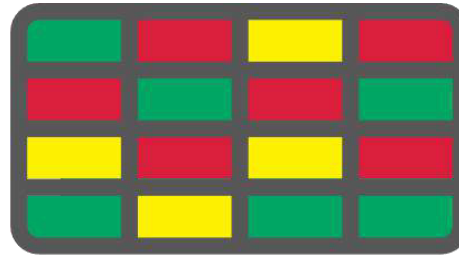


Pick by Light System

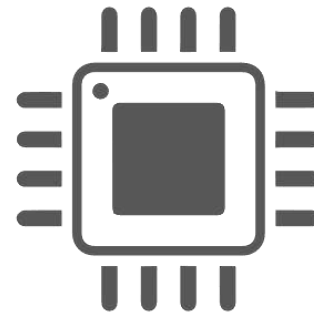
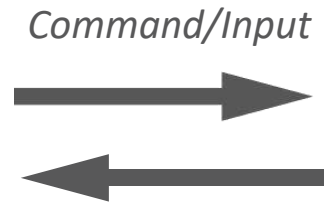
Uniquely Illuminable Component Storage

- Less **time** spent pulling inventory, more during assembly
- Easily avoid **shortages** via the Kanban system

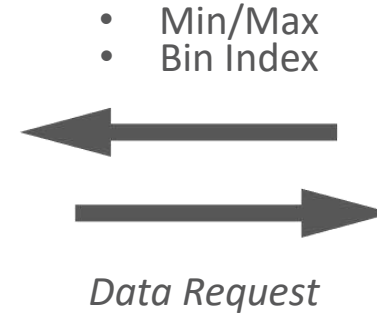
LED Manifestation of Location/Level Intuitive Pulling of Inventory



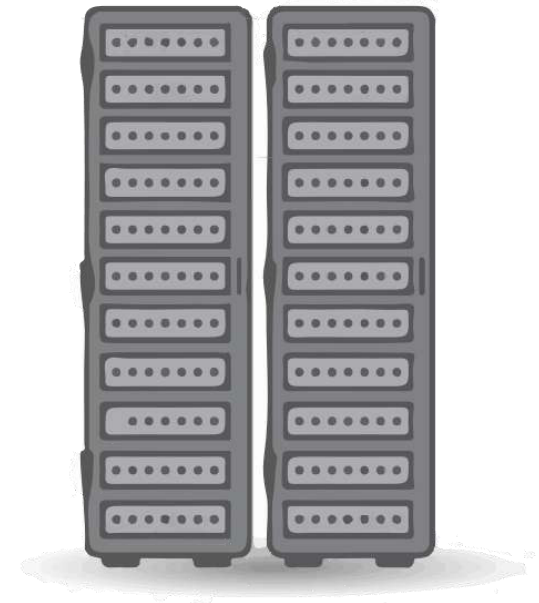
Android UI
(StratoCyt)



Embedded
Computer

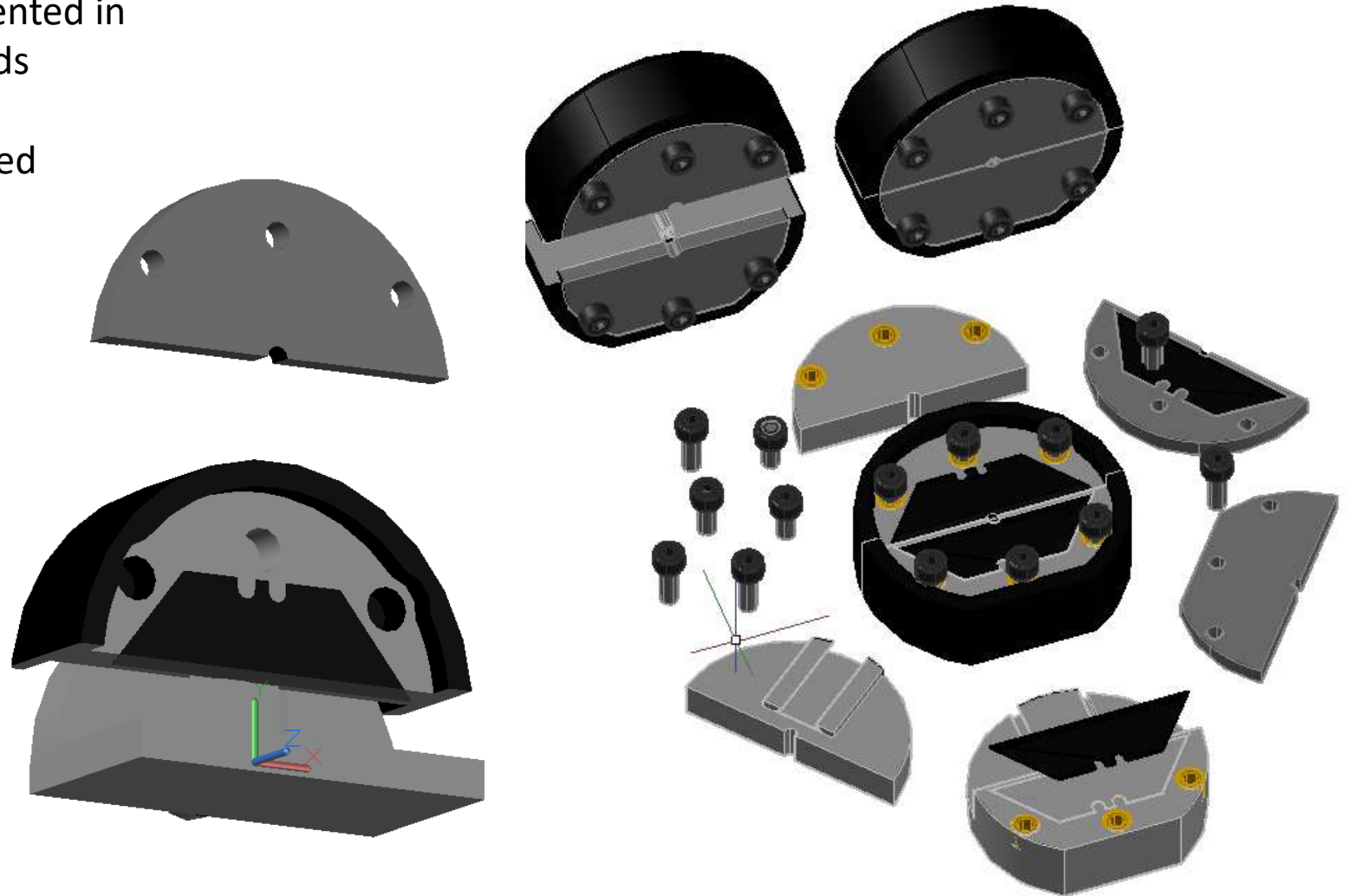


SQL Database

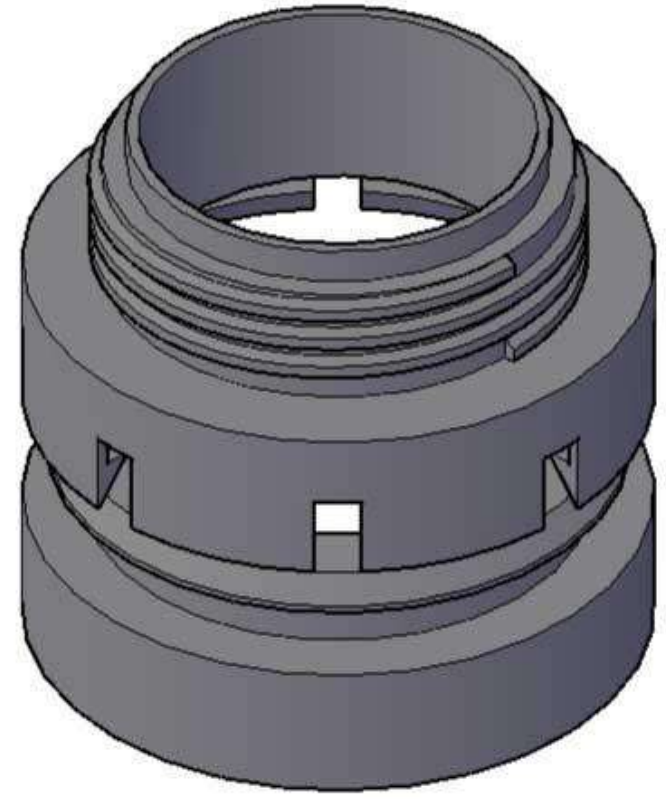
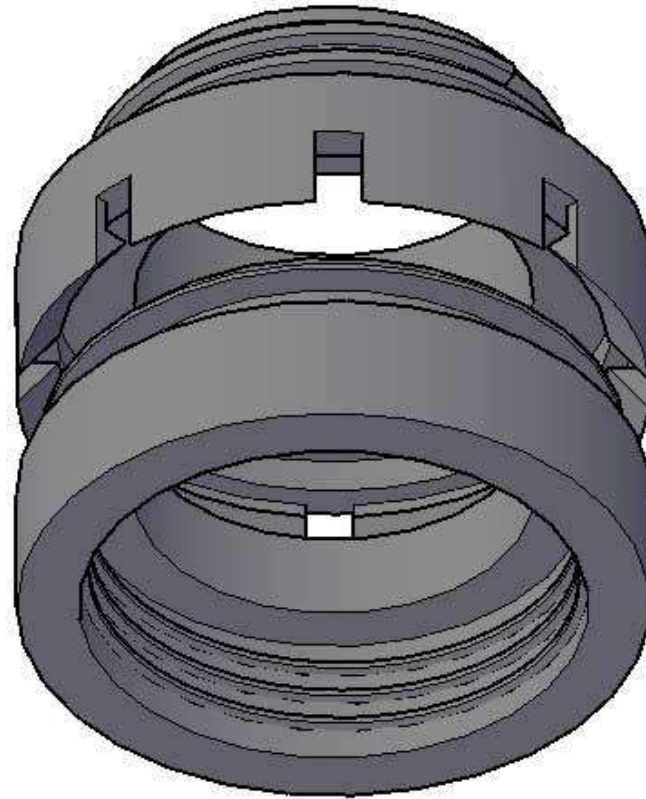
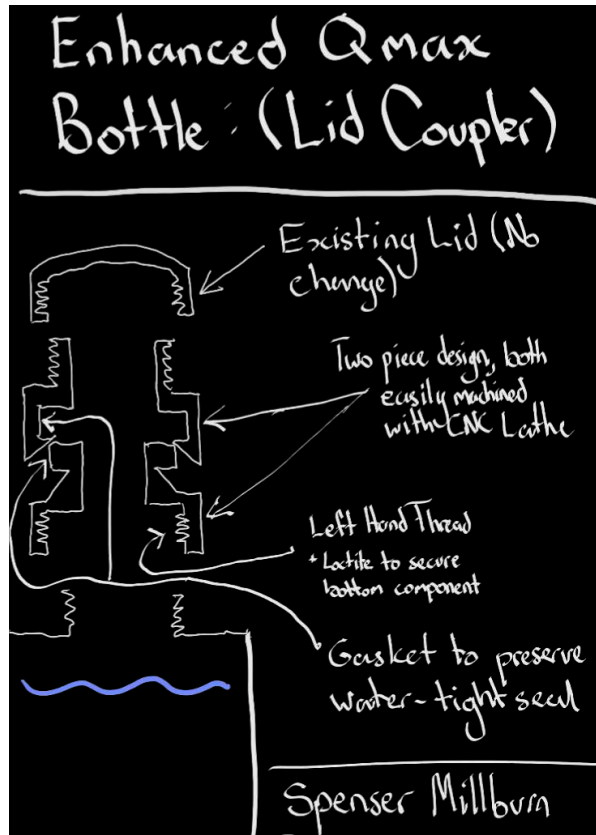


Microfluidics: Rework Jig

- Designed, fabricated, tested and implemented in 72hr window to meet production demands
- Effectively recovered USD \$20k in reworked material due to out of warranty vendor error



Qmax – Fluidics/Electrical Bearing`



Load Cell Calibration Jig

